

ADVANCED QUANTUM MECHANICS

Fundamentals of quantum mechanics



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LECTURE 4: Time independent perturbation theory

We first discuss the hydrogen atom orbitals and the important fact that, for a given principal quantum number n , all substates within this level are degenerate in energy. As shown in Fig. 1, each energy level exhibits an n^2 -fold degeneracy. This arises because for each $\ell = 0, \dots, n-1$ there are $(2\ell + 1)$ possible values of the magnetic quantum number m . Summing over all ℓ gives

$$\sum_{\ell=0}^{n-1} (2\ell + 1) = 2 \sum_{\ell=0}^{n-1} \ell + \sum_{\ell=0}^{n-1} 1 = 2 \frac{n(n-1)}{2} + n = n^2.$$

We then proceed to:

- introduce the mechanisms responsible for the splitting of electronic levels in the hydrogen atom,
- discuss the effects of the electronic and nuclear spins,
- present the framework of time-independent perturbation theory,
- and address the special case of degeneracies within time-independent perturbation theory.

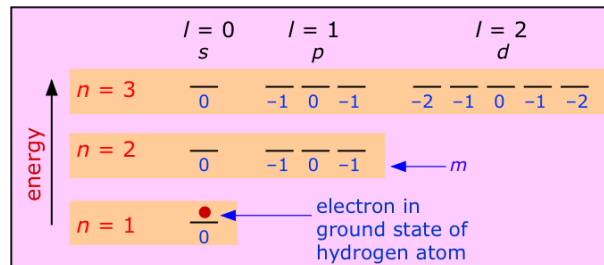


Figure 1: Three energy levels with their n^2 -fold degeneracy. This comes from summing the degeneracy of each ℓ state, which is $2\ell + 1$. In total this amounts to $1 + (2 \times 1 + 1) + (2 \times 2 + 1) + \dots$ up to $\ell = n - 1$, equalling n^2 .

The hydrogen atom - all sorts of corrections

An electronic orbital represents a spatially distributed, time-dependent charge density. As such, it can couple to external *electric* and *magnetic* fields—either static or time-varying. These couplings lift the degeneracy of atomic energy levels in different ways, depending on the specific form of the interaction.

In general, the interaction between a charge or current distribution and external fields is described by the Hamiltonian

$$\hat{H}_{\text{int}} = -\hat{\mathbf{d}} \cdot \mathbf{E} - \hat{\boldsymbol{\mu}} \cdot \mathbf{B},$$

where $\hat{\mathbf{d}} = -e \hat{\mathbf{r}}$ is the *electric dipole operator*, and $\hat{\boldsymbol{\mu}}$ is the *magnetic dipole operator*. Both moments may have *static* or *time-dependent* components, enabling coupling either to *static fields* or to *electromagnetic radiation*.

External and radiative effects

There are several external and quantum effects that lift degeneracies or shift energy levels:

(i) External classical fields

- **Zeeman effect (magnetic fields)** – Applying a magnetic field breaks rotational symmetry, lifting the degeneracy in the magnetic quantum number m . The interaction Hamiltonian between the magnetic moment and the field is

$$\hat{H}_Z = -\boldsymbol{\mu} \cdot \mathbf{B}.$$

For an electron with orbital angular momentum $\hat{\mathbf{L}}$, the magnetic moment is

$$\boldsymbol{\mu}_L = -\frac{e}{2m_e} \hat{\mathbf{L}}, \quad \Rightarrow \quad \hat{H}_Z = \frac{e}{2m_e} \hat{\mathbf{L}} \cdot \mathbf{B}.$$

For the $\hat{L}_z$ component the eigenvalues are $\hbar m$ and thus the eigenvalues of the electronic magnetic moment will be $m\mu_B$ where the Bohr magneton is

$$\mu_B = \frac{e\hbar}{2m_e}.$$

- **Stark effect (electric fields)** – An external electric field lifts degeneracies by coupling states with different ℓ and m . In a static field $\mathbf{E}$, the interaction Hamiltonian is

$$\hat{H}_S = -\hat{\mathbf{d}} \cdot \mathbf{E} = e \mathbf{E} \cdot \hat{\mathbf{r}},$$

where $\hat{\mathbf{d}} = -e \hat{\mathbf{r}}$ is the electric dipole operator. For

$$\mathbf{E} = E \hat{\mathbf{z}}: \quad \hat{H}_S = eE \hat{z} = eE \hat{r} \cos \theta,$$

and to compute the effect one writes

$$\langle n' \ell' m' | \hat{H}_S | n \ell m \rangle = eE \int_0^\infty r^3 dr R_{n' \ell'}(r) R_{n \ell}(r) \int d\Omega Y_{\ell' m'}^*(\Omega) \cos \theta Y_{\ell m}(\Omega).$$

This perturbation mixes states of opposite parity, producing energy shifts known as the Stark effect.

- (ii) **Quantum electrodynamic corrections** – The quantized electromagnetic field interacts with the electron, leading to observable effects such as the *Lamb shift* (radiative energy correction) and *spontaneous emission* (finite linewidth due to irreversible decay).

In addition there are direct and indirect consequences of relativity. These are owed to the fact that the Schrödinger equation is an approximation of the [Dirac equation](#)

$$\hat{H}_D = c \boldsymbol{\alpha} \cdot \hat{\mathbf{p}} + \beta mc^2 + V(\mathbf{r}) \mathbb{1}_4.$$

which we will discuss extensively in a later lecture. Here, $\boldsymbol{\alpha}$ and β are 4×4 Dirac matrices acting on the four-component spinor. In the non-relativistic limit, the Schrödinger equation is valid for $\langle \hat{\mathbf{p}}^2 \rangle / 2m \ll mc^2$.

Direct relativistic corrections

- (i) **Relativistic kinetic energy correction** — This fine-structure splitting scales approximately as α^2/n^3 , where α is the fine-structure constant. It originates from expanding the relativistic energy:

$$E = \sqrt{p^2 c^2 + m_e^2 c^4} \approx m_e c^2 + \frac{p^2}{2m_e} - \frac{p^4}{8m_e^3 c^2} + \dots$$

- (ii) **Darwin term** — another correction due to the rapid “Zitterbewegung” of the electron in the Dirac theory. It is proportional to $\nabla^2 V = \frac{e^2}{\epsilon_0} \delta(\mathbf{r})$ and it contributes only to s -states ($\ell = 0$).

Finally, the indirect consequences of the relativistic quantum mechanics are connected to the prediction of **spin** for electron and nucleus.

Fine and hyperfine structure

- (i) **Spin–orbit coupling** – The electron spin gives rise to an *intrinsic magnetic moment*

$$\boldsymbol{\mu}_S = -g_s \frac{e}{2m_e} \hat{\mathbf{S}}, \quad g_s \approx 2.$$

At the same time, the electron's orbital motion constitutes a circulating current, producing an *orbital magnetic moment*

$$\boldsymbol{\mu}_L = -\frac{e}{2m_e} \hat{\mathbf{L}}.$$

In the electron's rest frame, the electric field of the nucleus transforms into a magnetic field that interacts with the spin magnetic moment. The resulting interaction energy, known as the spin–orbit coupling, is

$$\hat{H}_{\text{so}} = \frac{1}{2m_e^2 c^2} \frac{1}{r} \frac{dV(r)}{dr} \hat{\mathbf{L}} \cdot \hat{\mathbf{S}}.$$

The total electronic angular momentum operator is defined as the sum of the orbital and spin contributions:

$$\hat{\mathbf{J}} = \hat{\mathbf{L}} + \hat{\mathbf{S}}.$$

The spin–orbit interaction lifts the degeneracy between states with different total angular momentum quantum numbers $j = \ell \pm \frac{1}{2}$, where the values $\pm \frac{1}{2}$ correspond to the eigenvalues of the spin projection operator $\hat{S}_z$ as the electron is a spin $\frac{1}{2}$ particle.

- (ii) **Hyperfine structure** – Beyond fine-structure effects, an even smaller splitting arises from the interaction between the *magnetic moment of the electron spin* and the *magnetic moment of the nucleus*. The nucleus, having spin $\hat{\mathbf{I}}$, possesses a magnetic moment

$$\boldsymbol{\mu}_I = g_I \frac{e}{2m_p} \hat{\mathbf{I}},$$

where g_I – nuclear g -factor, determined experimentally for each isotope. This couples to the electron spin magnetic moment $\boldsymbol{\mu}_S$ via

$$\hat{H}_{\text{hf}} = A \hat{\mathbf{I}} \cdot \hat{\mathbf{S}},$$

where A is the hyperfine coupling constant, depending on the overlap of the electronic wave function with the nucleus. The total atomic angular momentum is then

$$\hat{\mathbf{F}} = \hat{\mathbf{I}} + \hat{\mathbf{J}},$$

To summarize these different coupling and the usual names given

Type	Coupling	Scaling (for $Z = 1$)	Operator	Effect
Fine (Spin–orbit)	Electron spin $\leftrightarrow$ orbital motion	α^4	$\hat{\mathbf{L}} \cdot \hat{\mathbf{S}}$	Splitting between $j = \ell \pm \frac{1}{2}$
Hyperfine	Electron spin $\leftrightarrow$ nuclear spin	$\frac{m_e}{m_p} \alpha^4$	$\hat{\mathbf{I}} \cdot \hat{\mathbf{S}}$	Splitting between $F = I + J$

Table 1: Comparison of fine and hyperfine structure contributions for hydrogen ($Z = 1$).

Finally let's talk about nomenclature for atomic states. An atomic state that includes all angular momentum couplings is denoted by the term symbol

$$n^{2S+1}L_J \text{ or, including hyperfine coupling, } n^{2S+1}L_{J,F},$$

where:

- n — principal quantum number,
- L — orbital angular momentum (denoted by letters: $s, p, d, f, g, \dots$ for $\ell = 0, 1, 2, 3, 4, \dots$),
- S — total electron spin,
- J — total electronic angular momentum ($\hat{\mathbf{J}} = \hat{\mathbf{L}} + \hat{\mathbf{S}}$),
- F — total atomic angular momentum including the nucleus ($\hat{\mathbf{F}} = \hat{\mathbf{I}} + \hat{\mathbf{J}}$).

For example, the ground state of hydrogen (with $I = \frac{1}{2}$) is written as

$$1^2S_{1/2,F}, \quad F = 0, 1,$$

corresponding to the hyperfine doublet that produces the 21 cm line.

Time independent perturbation theory

Let's start with a Hamiltonian with only two states, the ground state $|g\rangle^{(0)}$ and the excited state $|e\rangle^{(0)}$, with energies $E_g^{(0)}$ and $E_e^{(0)}$, such that

$$\hat{H}_0 = E_g^{(0)} |g\rangle^{(0)}\langle g| + E_e^{(0)} |e\rangle^{(0)}\langle e|.$$

In matrix form (using the basis $\{|g\rangle^{(0)}, |e\rangle^{(0)}\}$),

$$\hat{H}_0 = \begin{pmatrix} E_g^{(0)} & 0 \\ 0 & E_e^{(0)} \end{pmatrix}.$$

Now we add a perturbation to this Hamiltonian,

$$\hat{H} = \hat{H}_0 + \lambda \hat{V},$$

and we ask: what are the new eigenstates $|g(\lambda)\rangle$ and $|e(\lambda)\rangle$ of the perturbed system?

We will vary the coupling parameter λ such that for $\lambda = 0$ we recover the unperturbed case ($|g\rangle = |g\rangle^{(0)}$, $|e\rangle = |e\rangle^{(0)}$), and for $\lambda = 1$ the perturbation is fully applied.

Ansatz and normalization condition

Let's focus on the $|e\rangle$ state and assume a series expansion in powers of the perturbation parameter λ . We expand both the energy and the state as power series in λ :

$$E_e = E_e^{(0)} + \lambda E_e^{(1)} + \lambda^2 E_e^{(2)} + \dots, \quad |e\rangle = |e^{(0)}\rangle + \lambda |e^{(1)}\rangle + \lambda^2 |e^{(2)}\rangle + \dots$$

In particular, we will focus on introducing the algorithm only up to second order for clarity. We will make a particular choice for the state corrections such that the total state remains normalized to the same order:

$$\begin{aligned} \langle e|e\rangle &= \langle e^{(0)}|e^{(0)}\rangle + \lambda(\langle e^{(1)}|e^{(0)}\rangle + \langle e^{(0)}|e^{(1)}\rangle) \\ &\quad + \lambda^2(\langle e^{(2)}|e^{(0)}\rangle + \langle e^{(0)}|e^{(2)}\rangle + \langle e^{(1)}|e^{(1)}\rangle) + \mathcal{O}(\lambda^3) = 1 + \mathcal{O}(\lambda^3). \end{aligned}$$

This condition is fulfilled by the following normalization choices:

$$\langle e^{(0)}|e^{(1)}\rangle = 0, \quad \langle e^{(0)}|e^{(2)}\rangle = -\frac{1}{2} \langle e^{(1)}|e^{(1)}\rangle.$$

Now we proceed to insert the expansions into the time-independent Schrödinger equation

$$(\hat{H}_0 + \lambda \hat{V}) |e\rangle = E_e |e\rangle$$

and act explicitly with $\hat{H}_0$ and $\hat{V}$ and expand both sides:

$$\underbrace{\hat{H}_0 |e^{(0)}\rangle}_{\lambda^0} + \lambda \underbrace{(\hat{H}_0 |e^{(1)}\rangle + \hat{V} |e^{(0)}\rangle)}_{\lambda^1} + \lambda^2 \underbrace{(\hat{H}_0 |e^{(2)}\rangle + \hat{V} |e^{(1)}\rangle)}_{\lambda^2} + \dots$$

$$= \underbrace{E_e^{(0)} |e^{(0)}\rangle}_{\lambda^0} + \lambda \underbrace{(E_e^{(0)} |e^{(1)}\rangle + E_e^{(1)} |e^{(0)}\rangle)}_{\lambda^1} + \lambda^2 \underbrace{(E_e^{(0)} |e^{(2)}\rangle + E_e^{(1)} |e^{(1)}\rangle + E_e^{(2)} |e^{(0)}\rangle)}_{\lambda^2} + \dots$$

Equating equal powers of λ leads to the following equations:

$$\text{Order } \lambda^0 : \hat{H}_0 |e^{(0)}\rangle = E_e^{(0)} |e^{(0)}\rangle, \quad (1a)$$

$$\text{Order } \lambda^1 : (\hat{H}_0 - E_e^{(0)}) |e^{(1)}\rangle = -\hat{V} |e^{(0)}\rangle + E_e^{(1)} |e^{(0)}\rangle, \quad (1b)$$

$$\text{Order } \lambda^2 : (\hat{H}_0 - E_e^{(0)}) |e^{(2)}\rangle = -\hat{V} |e^{(1)}\rangle + E_e^{(1)} |e^{(1)}\rangle + E_e^{(2)} |e^{(0)}\rangle. \quad (1c)$$

Order λ^1 We first solve the perturbed energy term and then move to find the perturbed eigenstate in this order.

Energy correction. - Project with $\langle e^{(0)}|$ (and use $\langle e^{(0)}| \hat{H}_0 = E_e^{(0)} \langle e^{(0)}|$) to get

$$\underbrace{\langle e^{(0)}| (\hat{H}_0 - E_e^{(0)}) |e^{(1)}\rangle}_{=0} = -\langle e^{(0)}| \hat{V} |e^{(0)}\rangle + E_e^{(1)} \underbrace{\langle e^{(0)}|e^{(0)}\rangle}_{=1},$$

hence

$$E_e^{(1)} = \langle e^{(0)}| \hat{V} |e^{(0)}\rangle = V_{ee}.$$

State correction. — By orthogonality, the first-order correction is proportional to the ground state:

$$|e^{(1)}\rangle = c_g^{(1)} |g^{(0)}\rangle.$$

Project with $\langle g^{(0)}|$:

$$\langle g^{(0)}| (\hat{H}_0 - E_e^{(0)}) |e^{(1)}\rangle = -\langle g^{(0)}| \hat{V} |e^{(0)}\rangle + E_e^{(1)} \langle g^{(0)}|e^{(0)}\rangle.$$

Since $\hat{H}_0 |g^{(0)}\rangle = E_g^{(0)} |g^{(0)}\rangle$ and $\langle g^{(0)}|e^{(0)}\rangle = 0$,

$$c_g^{(1)} (E_g^{(0)} - E_e^{(0)}) = -V_{ge} \Rightarrow c_g^{(1)} = \frac{V_{ge}}{\Delta E}.$$

Thus, to first order,

$$|e(\lambda)\rangle = |e^{(0)}\rangle + \lambda |e^{(1)}\rangle = |e^{(0)}\rangle + \lambda \frac{V_{ge}}{\Delta E} |g^{(0)}\rangle.$$

Order λ^2 Start from

$$(\hat{H}_0 - E_e^{(0)}) |e^{(2)}\rangle = -\hat{V} |e^{(1)}\rangle + E_e^{(1)} |e^{(1)}\rangle + E_e^{(2)} |e^{(0)}\rangle.$$

Energy correction. — Project with $\langle e^{(0)}|$ and use $\langle e^{(0)}|e^{(1)}\rangle = 0$:

$$E_e^{(2)} = \langle e^{(0)}| \hat{V} |e^{(1)}\rangle = \langle e^{(0)}| \hat{V} |g^{(0)}\rangle \frac{V_{ge}}{\Delta E} = \frac{|V_{ge}|^2}{\Delta E}.$$

State correction. — Write $|e^{(2)}\rangle = a |e^{(0)}\rangle + b |g^{(0)}\rangle$. Normalization gives

$$a = -\frac{1}{2} \langle e^{(1)}|e^{(1)}\rangle = -\frac{1}{2} \frac{|V_{ge}|^2}{\Delta E^2}.$$

Project the $\mathcal{O}(\lambda^2)$ equation with $\langle g^{(0)}|$:

$$(E_g^{(0)} - E_e^{(0)}) b = -\langle g^{(0)}| \hat{V} |e^{(1)}\rangle + E_e^{(1)} \langle g^{(0)}|e^{(1)}\rangle.$$

Using $|e^{(1)}\rangle = \frac{V_{ge}}{\Delta E} |g^{(0)}\rangle$ and $E_e^{(1)} = V_{ee}$,

$$\langle g^{(0)} | \hat{V} | e^{(1)} \rangle = \frac{V_{gg} V_{ge}}{\Delta E}, \quad \langle g^{(0)} | e^{(1)} \rangle = \frac{V_{ge}}{\Delta E}.$$

Hence

$$(E_g^{(0)} - E_e^{(0)}) b = \frac{(V_{ee} - V_{gg}) V_{ge}}{\Delta E} \Rightarrow b = \frac{(V_{gg} - V_{ee}) V_{ge}}{\Delta E^2}.$$

Therefore,

$$|e^{(2)}\rangle = \left[-\frac{1}{2} \frac{|V_{ge}|^2}{\Delta E^2} \right] |e^{(0)}\rangle + \left[\frac{(V_{gg} - V_{ee}) V_{ge}}{\Delta E^2} \right] |g^{(0)}\rangle.$$

Let's make the simplifying notation $V_{ge} = \langle g^{(0)} | \hat{V} | e^{(0)} \rangle$. Collecting all results, we can now summarize the perturbative expansion up to second order:

Two-level perturbation summary up to second order (for excited state)

Notation:

$$\Delta E := E_e^{(0)} - E_g^{(0)}, \quad V_{ee} := \langle e^{(0)} | \hat{V} | e^{(0)} \rangle, \quad V_{gg} := \langle g^{(0)} | \hat{V} | g^{(0)} \rangle, \quad V_{ge} := \langle g^{(0)} | \hat{V} | e^{(0)} \rangle, \quad V_{eg} = V_{ge}^*.$$

Normalization condition $\langle e^{(0)} | e \rangle = 1$

Energy

$$E_e = E_e^{(0)} + \lambda V_{ee} + \lambda^2 \frac{|V_{ge}|^2}{\Delta E} + \mathcal{O}(\lambda^3).$$

State vector

$$|e\rangle = |e^{(0)}\rangle + \lambda |e^{(1)}\rangle + \lambda^2 |e^{(2)}\rangle + \mathcal{O}(\lambda^3),$$

with the first two orders

$$|e^{(1)}\rangle = |g^{(0)}\rangle \frac{V_{ge}}{\Delta E},$$

$$|e^{(2)}\rangle = \left[-\frac{1}{2} \frac{|V_{ge}|^2}{\Delta E^2} \right] |e^{(0)}\rangle + \left[\frac{(V_{gg} - V_{ee}) V_{ge}}{\Delta E^2} \right] |g^{(0)}\rangle.$$

As a check notice that for $\lambda = 0$ we get the unperturbed states and energies, as expected. Slowly increasing λ is the smart way to proceed to make sure the perturbation theory works, until getting to the point where $\lambda = 1$ gives the perturbed eigenproblem fully.

Of course this procedure can easily be extended to include any numbers of additional states.

Degenerate perturbation theory (perturbation theory within the degenerate manifold)

Let us again restrict to two states as before, but now assume that they have the same energy. Clearly, the procedure used previously fails: the first-order correction involves denominators $(E_0 - E_n^{(0)})^{-1}$. When two or more unperturbed states share the same energy, these denominators vanish, and the expansion becomes ill-defined. Degenerate perturbation theory avoids this problem by [first diagonalizing the perturbation within the degenerate subspace](#).

We assume here that $E_g^{(0)} = E_e^{(0)} = E_0$. Notice that any linear combination of these two states,

$$|\psi^{(0)}\rangle = c_g |g^{(0)}\rangle + c_e |e^{(0)}\rangle,$$

is itself an eigenstate of $\hat{H}_0$ with eigenvalue E_0 . This means that $|g^{(0)}\rangle$ and $|e^{(0)}\rangle$ are not uniquely defined eigenstates within the degenerate subspace. Consequently, the perturbed states are not simply perturbations of $|g^{(0)}\rangle$ or $|e^{(0)}\rangle$, but rather new superpositions that diagonalize the perturbation operator $\hat{V}$ within this degenerate manifold. The task here becomes very different than before. The key idea is that within the degenerate subspace, $\hat{H}_0$ acts as $E_0 \mathbb{1}$, so the

unperturbed eigenstates are not unique. The physically correct zeroth-order states are the linear combinations that also diagonalize the perturbation $\hat{V}$ restricted to this subspace.

We therefore expand both the state vector and the energy as power series in the perturbation strength:

$$|\psi\rangle = |\psi^{(0)}\rangle + \lambda |\psi^{(1)}\rangle + \lambda^2 |\psi^{(2)}\rangle + \dots, \quad E = E_0 + \lambda E^{(1)} + \lambda^2 E^{(2)} + \dots,$$

and insert these into the stationary Schrödinger equation,

$$(\hat{H}_0 + \lambda \hat{V}) |\psi\rangle = E |\psi\rangle.$$

Grouping leads to

$$\begin{aligned} & \underbrace{\hat{H}_0 |\psi^{(0)}\rangle}_{\lambda^0} + \lambda \underbrace{(\hat{H}_0 |\psi^{(1)}\rangle + \hat{V} |\psi^{(0)}\rangle)}_{\lambda^1} + \lambda^2 \underbrace{(\hat{H}_0 |\psi^{(2)}\rangle + \hat{V} |\psi^{(1)}\rangle)}_{\lambda^2} + \dots \\ &= \underbrace{E_0 |\psi^{(0)}\rangle}_{\lambda^0} + \lambda \underbrace{(E_0 |\psi^{(1)}\rangle + E^{(1)} |\psi^{(0)}\rangle)}_{\lambda^1} + \lambda^2 \underbrace{(E_0 |\psi^{(2)}\rangle + E^{(1)} |\psi^{(1)}\rangle + E^{(2)} |\psi^{(0)}\rangle)}_{\lambda^2} + \dots \end{aligned}$$

$$\lambda^0: \quad \hat{H}_0 |\psi^{(0)}\rangle = E_0 |\psi^{(0)}\rangle. \quad (2a)$$

$$\lambda^1: \quad (\hat{H}_0 - E_0) |\psi^{(1)}\rangle = -\hat{V} |\psi^{(0)}\rangle + E^{(1)} |\psi^{(0)}\rangle. \quad (2b)$$

$$\lambda^2: \quad (\hat{H}_0 - E_0) |\psi^{(2)}\rangle = -\hat{V} |\psi^{(1)}\rangle + E^{(1)} |\psi^{(1)}\rangle + E^{(2)} |\psi^{(0)}\rangle. \quad (2c)$$

Order λ^1 - Acting with $\langle g^{(0)}|$ and $\langle e^{(0)}|$ gives

$$0 = -(V_{gg}c_g + V_{ge}c_e) + E^{(1)}c_g, \quad (3a)$$

$$0 = -(V_{eg}c_g + V_{ee}c_e) + E^{(1)}c_e, \quad (3b)$$

where $V_{\alpha\beta} = \langle \alpha^{(0)} | \hat{V} | \beta^{(0)} \rangle$, or in matrix form

$$\begin{pmatrix} V_{gg} - E^{(1)} & V_{ge} \\ V_{eg} & V_{ee} - E^{(1)} \end{pmatrix} \begin{pmatrix} c_g \\ c_e \end{pmatrix} = 0.$$

Nontrivial coefficients $(c_g, c_e) \neq (0, 0)$ only exist in the case where the determinant vanishes:

$$(V_{gg} - E^{(1)})(V_{ee} - E^{(1)}) - V_{ge}V_{eg} = 0.$$

Solving for $E^{(1)}$ gives the first-order energies

$$E_{\pm}^{(1)} = \frac{V_{gg} + V_{ee}}{2} \pm \sqrt{\left(\frac{V_{gg} - V_{ee}}{2}\right)^2 + |V_{ge}|^2}.$$

To find the ket state corrections in this order, we solve the first row of the linear system,

$$(V_{gg} - E_{\pm}^{(1)})c_g + V_{ge}c_e = 0 \quad \Rightarrow \quad \frac{c_e}{c_g} = -\frac{V_{gg} - E_{\pm}^{(1)}}{V_{ge}}.$$

After normalization,

$$|\psi_{\pm}^{(0)}\rangle = \frac{1}{\sqrt{1 + \left|\frac{V_{gg} - E_{\pm}^{(1)}}{V_{ge}}\right|^2}} \left(|g^{(0)}\rangle - \frac{V_{gg} - E_{\pm}^{(1)}}{V_{ge}} |e^{(0)}\rangle \right).$$

Substituting back into the λ^1 equation,

$$(\hat{H}_0 - E_0) |\psi^{(1)}\rangle = -\hat{V} |\psi_{\pm}^{(0)}\rangle + E_{\pm}^{(1)} |\psi_{\pm}^{(0)}\rangle.$$

By construction, the right-hand side is zero, hence

$$(\hat{H}_0 - E_0) |\psi_{\pm}^{(1)}\rangle = 0.$$

Since the two-state space is the whole Hilbert space, this implies

$$|\psi_{\pm}^{(1)}\rangle = 0.$$

The degeneracy is therefore lifted by the perturbation: the two states $|\psi_+^{(0)}\rangle$ and $|\psi_-^{(0)}\rangle$ acquire distinct first-order energies $E_+^{(1)}$ and $E_-^{(1)}$, separated by $2\sqrt{\left(\frac{V_{gg}-V_{ee}}{2}\right)^2 + |V_{ge}|^2}$. This splitting is often referred to as the level repulsion or lifting of degeneracy.

Note that this procedure generalizes naturally to any dimension of the degenerate subspace: one must diagonalize the perturbation matrix $V_{\alpha\beta} = \langle\alpha^{(0)}|\hat{V}|\beta^{(0)}\rangle$ within that subspace.